

DATA ANALYTICS

(Module 1)

DATA EXPLORATION

EXCEL ASSIGNMENT-1

DATA EXPLORATION:

Basic data exploration and analysis in Microsoft Excel. The dataset contains information about different products, including their Product IDs, Product Name, Brand Name, Quantity, Categories, and Prices.

The goal is to build foundational skills in data exploration, logical functions, conditional aggregation, and text manipulation using Excel.

Basic Data Exploration

1. Sum, Count, and Average:

SUM:

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	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	MID	RIGHT	SUMIF	COUNTIF										
2	JAN	US	8050	11										
3	FEB	US	7050	11										
4	MAR	US	7050	10										
5	APR	US	7050	10		SUM	Total price of all products		10100					
6	MAY	US	6150	10		COUNT	No. of products exist in the dat		34					
7	JUN	UK	6150	9		AVERAGE	Average price of the products		297.0588235					
8	JUL	UK	5950	9		MIN	minimum price among all proi		30					
9	AUG	UK	5950	8		MAX	Maximum price among all pro		1000					
10	SEP	UK	5950	7		IF FUNCTIONS:								
11	OCT	UK	5450	7		SUMIF	Electronics price		\$ 7,100.00					
12	JUN	IN	5450	7		COUNTIF	less than \$100.		\$ 11.00					
13	NOV	AU	4500	7										
14	DEC	DE	4500	6										
15	FEB	CA	4500	6										
16	APR	ES	4350	6										
17	AUG	CA	4100	6										
18	AUG	CN	4100	5										
19	JAN	IT	3940	5										
20	MAR	UK	2960	5										

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COUNT:

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Q6: =COUNTA(D2:D35)

	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	MID	RIGHT	SUMIF	COUNTIF										
2	JAN	US	8050	11										
3	FEB	US	7050	11										
4	MAR	US	7050	10										
5	APR	US	7050	10		SUM	Total price of all products		10100					
6	MAY	US	6150	10		COUNT	No. of products exist in the data		34					
7	JUN	UK	6150	9		AVERAGE	Average price of the products		297.0588235					
8	JUL	UK	5950	9		MIN	minimum price among all products		30					
9	AUG	UK	5950	8		MAX	Maximum price among all products		1000					
10	SEP	UK	5950	7		IF FUNCTIONS:								
11	OCT	UK	5450	7		SUMIF	Electronics price		\$ 7,100.00					
12	JUN	IN	5450	7		COUNTIF	less than \$100.		\$ 11.00					
13	NOV	AU	4500	7										
14	DEC	DE	4500	6										
15	FEB	CA	4500	6										
16	APR	ES	4350	6										
17	AUG	CA	4100	6										
18	AUG	CN	4100	5										
19	JAN	IT	3940	5										
20	MAR	UK	2960	5										

Dataset

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AVERAGE:

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Q7: =AVERAGE(D2:D35)

	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	MID	RIGHT	SUMIF	COUNTIF										
2	JAN	US	8050	11										
3	FEB	US	7050	11										
4	MAR	US	7050	10										
5	APR	US	7050	10		SUM	Total price of all products		10100					
6	MAY	US	6150	10		COUNT	No. of products exist in the data		34					
7	JUN	UK	6150	9		AVERAGE	Average price of the products		297.0588235					
8	JUL	UK	5950	9		MIN	minimum price among all products		30					
9	AUG	UK	5950	8		MAX	Maximum price among all products		1000					
10	SEP	UK	5950	7		IF FUNCTIONS:								
11	OCT	UK	5450	7		SUMIF	Electronics price		\$ 7,100.00					
12	JUN	IN	5450	7		COUNTIF	less than \$100.		\$ 11.00					
13	NOV	AU	4500	7										
14	DEC	DE	4500	6										
15	FEB	CA	4500	6										
16	APR	ES	4350	6										
17	AUG	CA	4100	6										
18	AUG	CN	4100	5										
19	JAN	IT	3940	5										
20	MAR	UK	2960	5										

Dataset

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2. Minimum and Maximum Values

Using Excel functions:

MINIMUM:

The screenshot shows the Microsoft Excel interface with the 'FORMULAS' tab selected. The formula bar displays '=MIN(D:D)'. The worksheet contains a table with columns: MID, RIGHT, SUMIF, COUNTIF, and a summary section. The summary section includes rows for SUM, COUNT, AVERAGE, MIN, and MAX. The MIN function is applied to the 'SUMIF' column, resulting in the value 30.

	MID	RIGHT	SUMIF	COUNTIF
1	JAN	US	8050	11
2	FEB	US	7050	11
3	MAR	US	7050	10
4	APR	US	7050	10
5	MAY	US	6150	10
6	JUN	UK	6150	9
7	JUL	UK	5950	9
8	AUG	UK	5950	8
9	SEP	UK	5950	7
10	OCT	UK	5450	7
11	JUN	IN	5450	7
12	NOV	AU	4500	7
13	DEC	DE	4500	6
14	FEB	CA	4500	6
15	APR	ES	4350	6
16	AUG	CA	4100	6
17	AUG	CN	4100	5
18	JAN	IT	3940	5
19	MAR	UK	2960	5

Summary Section:

	SUM	COUNT	AVERAGE	MIN	MAX
Total price of all products	10100				
No. of products exist in the data		34			
Average price of the products		297.0588235			
minimum price among all products				30	
Maximum price among all products					1000

MAXIMUM:

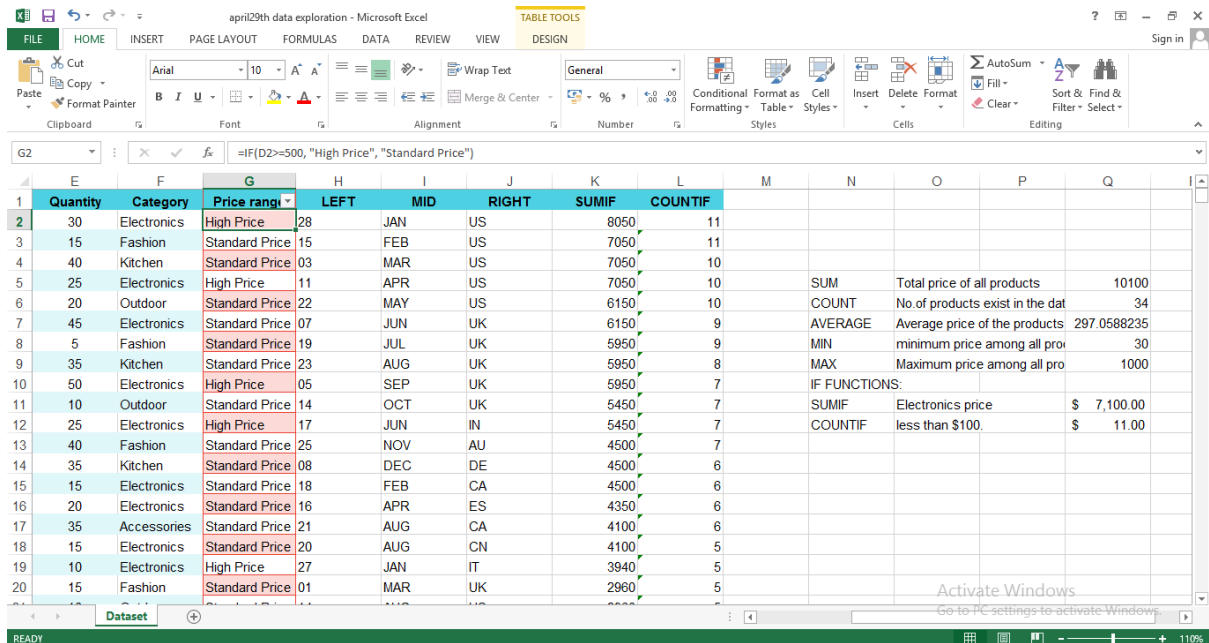
The screenshot shows the Microsoft Excel interface with the 'FORMULAS' tab selected. The formula bar displays '=MAX(D2:D35)'. The worksheet contains the same table as the previous screenshot. The MAX function is applied to the 'SUMIF' column, resulting in the value 1000.

	MID	RIGHT	SUMIF	COUNTIF
1	JAN	US	8050	11
2	FEB	US	7050	11
3	MAR	US	7050	10
4	APR	US	7050	10
5	MAY	US	6150	10
6	JUN	UK	6150	9
7	JUL	UK	5950	9
8	AUG	UK	5950	8
9	SEP	UK	5950	7
10	OCT	UK	5450	7
11	JUN	IN	5450	7
12	NOV	AU	4500	7
13	DEC	DE	4500	6
14	FEB	CA	4500	6
15	APR	ES	4350	6
16	AUG	CA	4100	6
17	AUG	CN	4100	5
18	JAN	IT	3940	5
19	MAR	UK	2960	5

Summary Section:

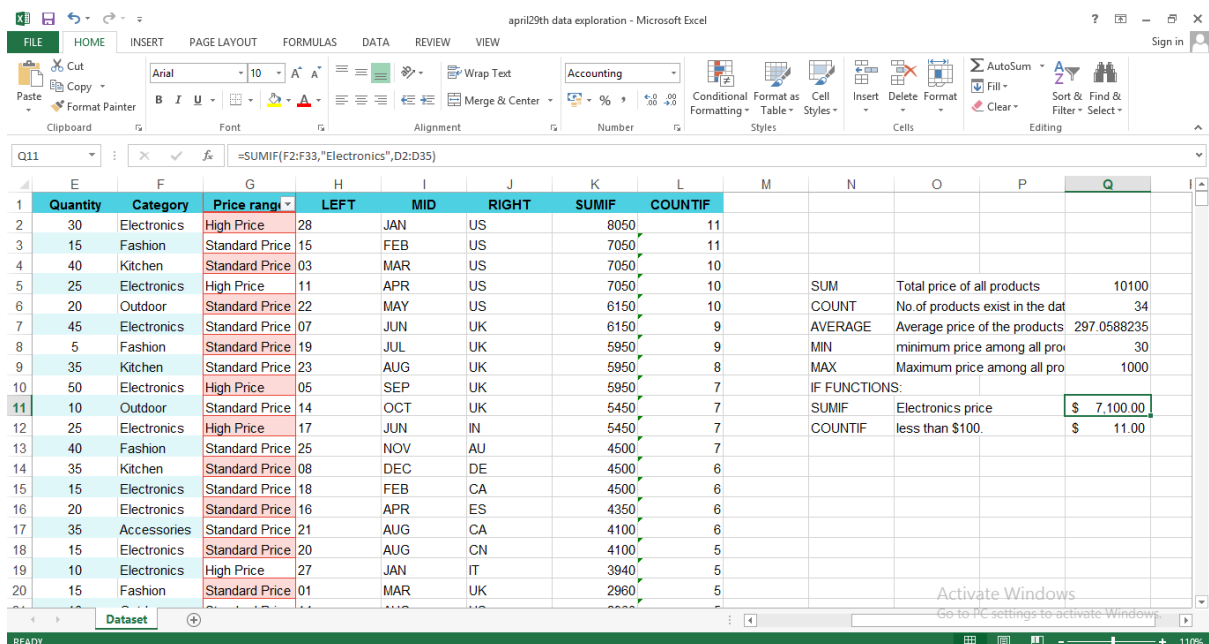
	SUM	COUNT	AVERAGE	MIN	MAX
Total price of all products	10100				
No. of products exist in the data		34			
Average price of the products		297.0588235			
minimum price among all products				30	
Maximum price among all products					1000

3. Logical Function – IF



4. Conditional Functions – SUMIF and COUNTIF

SUMIF:



COUNTIF:

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Clipboard Font Alignment Number Styles Cells Editing

Q12 =COUNTIF(D:D,"<100")

	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Quantity	Category	Price range	LEFT	MID	RIGHT	SUMIF	COUNTIF					
2	30	Electronics	High Price	28	JAN	US	8050	11					
3	15	Fashion	Standard Price	15	FEB	US	7050	11					
4	40	Kitchen	Standard Price	03	MAR	US	7050	10					
5	25	Electronics	High Price	11	APR	US	7050	10		SUM	Total price of all products		10100
6	20	Outdoor	Standard Price	22	MAY	US	6150	10		COUNT	No. of products exist in the dat		34
7	45	Electronics	Standard Price	07	JUN	UK	6150	9		AVERAGE	Average price of the products		297.0588235
8	5	Fashion	Standard Price	19	JUL	UK	5950	9		MIN	minimum price among all pro		30
9	35	Kitchen	Standard Price	23	AUG	UK	5950	8		MAX	Maximum price among all pro		1000
10	50	Electronics	High Price	05	SEP	UK	5950	7		IF FUNCTIONS:			
11	10	Outdoor	Standard Price	14	OCT	UK	5450	7		SUMIF	Electronics price		\$ 7,100.00
12	25	Electronics	High Price	17	JUN	IN	5450	7		COUNTIF	less than \$100.		\$ 11.00
13	40	Fashion	Standard Price	25	NOV	AU	4500	7					
14	35	Kitchen	Standard Price	08	DEC	DE	4500	6					
15	15	Electronics	Standard Price	18	FEB	CA	4500	6					
16	20	Electronics	Standard Price	16	APR	ES	4350	6					
17	35	Accessories	Standard Price	21	AUG	CA	4100	6					
18	15	Electronics	Standard Price	20	AUG	CN	4100	5					
19	10	Electronics	High Price	27	JAN	IT	3940	5					
20	15	Fashion	Standard Price	01	MAR	UK	2980	5					

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5. Text Formatting – LEFT, RIGHT, MID:

LEFT :

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Table Name: Table2

Properties Tools External Table Data

Table Style Options

Table Styles

H2 =LEFT(A2,2)

	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Quantity	Category	Price range	DAY	MONTH	COUNTRY	COD	SUMIF	COUNTIF				
2	30	Electronics	High Price	28	JAN	US		8050	11				
3	15	Fashion	Standard Price	15	FEB	US		7050	11				
4	40	Kitchen	Standard Price	03	MAR	US		7050	10				
5	25	Electronics	High Price	11	APR	US		7050	10		SUM	Total price of all products	10100
6	20	Outdoor	Standard Price	22	MAY	US		6150	10		COUNT	No. of products exist in the dat	34
7	45	Electronics	Standard Price	07	JUN	UK		6150	9		AVERAGE	Average price of the products	297.0588235
8	5	Fashion	Standard Price	19	JUL	UK		5950	9		MIN	minimum price among all pro	30
9	35	Kitchen	Standard Price	23	AUG	UK		5950	8		MAX	Maximum price among all pro	1000
10	50	Electronics	High Price	05	SEP	UK		5950	7		IF FUNCTIONS:		
11	10	Outdoor	Standard Price	14	OCT	UK		5450	7		SUMIF	Electronics price	\$ 7,100.00
12	25	Electronics	High Price	17	JUN	IN		5450	7		COUNTIF	less than \$100.	\$ 11.00
13	40	Fashion	Standard Price	25	NOV	AU		4500	7				
14	35	Kitchen	Standard Price	08	DEC	DE		4500	6				
15	15	Electronics	Standard Price	18	FEB	CA		4500	6				
16	20	Electronics	Standard Price	16	APR	ES		4350	6				
17	35	Accessories	Standard Price	21	AUG	CA		4100	6				
18	15	Electronics	Standard Price	20	AUG	CN		4100	5				
19	10	Electronics	High Price	27	JAN	IT		3940	5				
20	15	Fashion	Standard Price	01	MAR	UK		2980	5				

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MID:

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FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW DESIGN

Table Name: Table3

Summarize with PivotTable
Remove Duplicates
Convert to Range

Insert Slicer
Export
Refresh
Open in Browser
Unlink

Properties
External Table Data

TABLE TOOLS
DESIGN
Header Row
Total Row
Banded Rows
First Column
Last Column
Banded Columns
Filter Button

Table Styles

Properties Tools External Table Data Table Style Options Table Styles

I2 =MID(A2,4,3)

	E	F	G	H	I	J	K	L	M	N	O	P	Q
	Quantity	Category	Price range	DAY	MONTH	COUNTRYCOD	SUMIF	COUNTIF					
1	30	Electronics	High Price	28	JAN	US	8050	11					
2	15	Fashion	Standard Price	15	FEB	US	7050	11					
3	40	Kitchen	Standard Price	03	MAR	US	7050	10					
4	25	Electronics	High Price	11	APR	US	7050	10					
5	20	Outdoor	Standard Price	22	MAY	US	6150	10		SUM	Total price of all products		10100
6	45	Electronics	Standard Price	07	JUN	UK	6150	9		COUNT	No of products exist in the dat		34
7	5	Fashion	Standard Price	19	JUL	UK	5950	9		AVERAGE	Average price of the products		297.0588235
8	35	Kitchen	Standard Price	23	AUG	UK	5950	8		MIN	minimum price among all pro		30
9	50	Electronics	High Price	05	SEP	UK	5950	7		MAX	Maximum price among all pro		1000
10	10	Outdoor	Standard Price	14	OCT	UK	5450	7		IF FUNCTIONS:			
11	25	Electronics	High Price	17	JUN	IN	5450	7		SUMIF	Electronics price		\$ 7,100.00
12	40	Fashion	Standard Price	25	NOV	AU	4500	7		COUNTIF	less than \$100.		\$ 11.00
13	35	Kitchen	Standard Price	08	DEC	DE	4500	6					
14	15	Electronics	Standard Price	18	FEB	CA	4500	6					
15	20	Electronics	Standard Price	16	APR	ES	4350	6					
16	35	Accessories	Standard Price	21	AUG	CA	4100	6					
17	15	Electronics	Standard Price	20	AUG	CN	4100	5					
18	10	Electronics	High Price	27	JAN	IT	3940	5					
19	15	Fashion	Standard Price	01	MAR	UK	2960	5					

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RIGHT:

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FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW DESIGN

Table Name: Table4

Summarize with PivotTable
Remove Duplicates
Convert to Range

Insert Slicer
Export
Refresh
Open in Browser
Unlink

Properties
External Table Data

TABLE TOOLS
DESIGN
Header Row
Total Row
Banded Rows
First Column
Last Column
Banded Columns
Filter Button

Table Styles

Properties Tools External Table Data Table Style Options Table Styles

J2 =RIGHT(A2,2)

	E	F	G	H	I	J	K	L	M	N	O	P	Q
	Quantity	Category	Price range	DAY	MONTH	COUNTRYCODE	SUMIF	COUNTIF					
1	30	Electronics	High Price	28	JAN	US	8050	11					
2	15	Fashion	Standard Price	15	FEB	US	7050	11					
3	40	Kitchen	Standard Price	03	MAR	US	7050	10					
4	25	Electronics	High Price	11	APR	US	7050	10					
5	20	Outdoor	Standard Price	22	MAY	US	6150	10		SUM	Total price of all products		10100
6	45	Electronics	Standard Price	07	JUN	UK	6150	9		COUNT	No of products exist in the dat		34
7	5	Fashion	Standard Price	19	JUL	UK	5950	9		AVERAGE	Average price of the products		297.0588235
8	35	Kitchen	Standard Price	23	AUG	UK	5950	8		MIN	minimum price among all pro		30
9	50	Electronics	High Price	05	SEP	UK	5950	7		MAX	Maximum price among all pro		1000
10	10	Outdoor	Standard Price	14	OCT	UK	5450	7		IF FUNCTIONS:			
11	25	Electronics	High Price	17	JUN	IN	5450	7		SUMIF	Electronics price		\$ 7,100.00
12	40	Fashion	Standard Price	25	NOV	AU	4500	7		COUNTIF	less than \$100.		\$ 11.00
13	35	Kitchen	Standard Price	08	DEC	DE	4500	6					
14	15	Electronics	Standard Price	18	FEB	CA	4500	6					
15	20	Electronics	Standard Price	16	APR	ES	4350	6					
16	35	Accessories	Standard Price	21	AUG	CA	4100	6					
17	15	Electronics	Standard Price	20	AUG	CN	4100	5					
18	10	Electronics	High Price	27	JAN	IT	3940	5					
19	15	Fashion	Standard Price	01	MAR	UK	2960	5					

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Dataset

RESULT ANALYSIS:

Product analysis and results of exploring the dataset given:

- **Total Inventory Value:** The sum of all product prices is **\$10,100**.
- **Product Volume:** There are **34** individual products listed in the dataset.
- **Price Range:** Prices vary significantly, from a minimum of **\$30** to a maximum of **\$1,000**.
- **Average Price:** The average cost per product is approximately **\$297.0**
- **Electronics Dominance:** The SUMIF function shows that "Electronics" account for **\$7,100**, representing roughly **70%** of the total inventory value.
- **Affordable Options:** There are **11** products priced under \$100
- The summary effectively utilizes core Excel functions to analyze the data:
- **Basic Aggregation:** SUM, COUNT, and AVERAGE.
- **Range Analysis:** MIN and MAX.
- **Conditional Analysis:** SUMIF (for category-specific totals) and COUNTIF (for price bracket distribution).